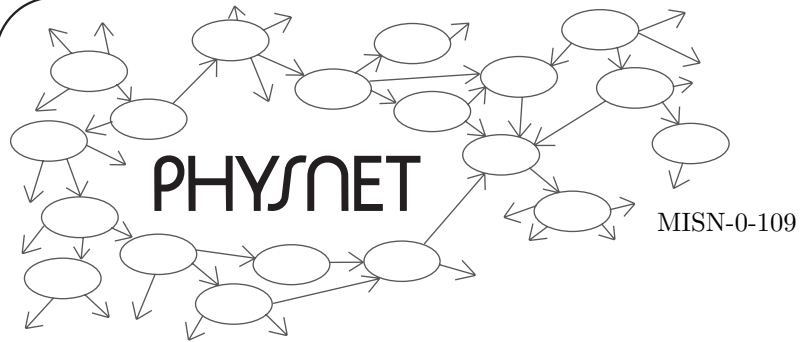
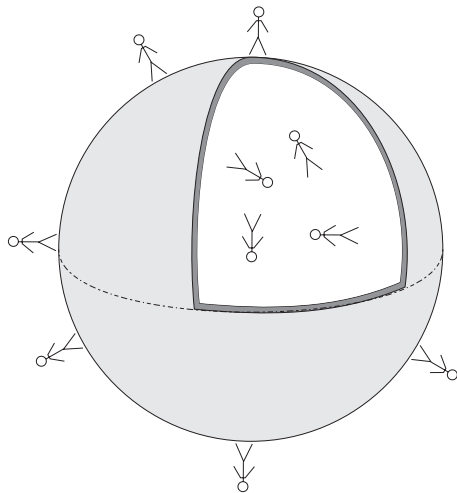




THE GRAVITATIONAL FIELD OUTSIDE A HOMOGENEOUS SPHERICAL MASS



THE GRAVITATIONAL FIELD OUTSIDE A HOMOGENEOUS SPHERICAL MASS

by

P. Signell and J. S. Kovacs

1. Introduction	1
2. Study Material	1
Acknowledgments	2

Title: **The Gravitational Field Outside a Homogeneous Spherical Mass**

Author: P. Signell and J. S Kovacs, Michigan State University

Version: 2/1/2000

Evaluation: Stage 0

Length: 1 hr; 8 pages

Input Skills:

1. Explain the meanings of the symbols in the Law of Universal Gravitation (MISN-0-101).
2. Calculate the gravitational potential due to a distribution of sources (MISN-0-107).
3. Calculate the gravitational field from the gravitational potential (MISN-0-108).

Output Skills (Knowledge):

- K1. Derive the gravitational field and potential outside of hollow and solid spheres, hence justify use of the point-mass Law of Universal Gravitation near a mass that has spherical symmetry.

External Resources (Required):

1. M. Alonso and E. J. Finn, *Physics*, Addison-Wesley, Reading (1970); For availability, see this module's *Local Guide*.

THIS IS A DEVELOPMENTAL-STAGE PUBLICATION
OF PROJECT PHYSNET

The goal of our project is to assist a network of educators and scientists in transferring physics from one person to another. We support manuscript processing and distribution, along with communication and information systems. We also work with employers to identify basic scientific skills as well as physics topics that are needed in science and technology. A number of our publications are aimed at assisting users in acquiring such skills.

Our publications are designed: (i) to be updated quickly in response to field tests and new scientific developments; (ii) to be used in both classroom and professional settings; (iii) to show the prerequisite dependencies existing among the various chunks of physics knowledge and skill, as a guide both to mental organization and to use of the materials; and (iv) to be adapted quickly to specific user needs ranging from single-skill instruction to complete custom textbooks.

New authors, reviewers and field testers are welcome.

PROJECT STAFF

Andrew Schnepf	Webmaster
Eugene Kales	Graphics
Peter Signell	Project Director

ADVISORY COMMITTEE

D. Alan Bromley	Yale University
E. Leonard Jossem	The Ohio State University
A. A. Strassenburg	S. U. N. Y., Stony Brook

Views expressed in a module are those of the module author(s) and are not necessarily those of other project participants.

© 2001, Peter Signell for Project PHYSNET, Physics-Astronomy Bldg., Mich. State Univ., E. Lansing, MI 48824; (517) 355-3784. For our liberal use policies see:

<http://www.physnet.org/home/modules/license.html>.

THE GRAVITATIONAL FIELD OUTSIDE A HOMOGENEOUS SPHERICAL MASS

by

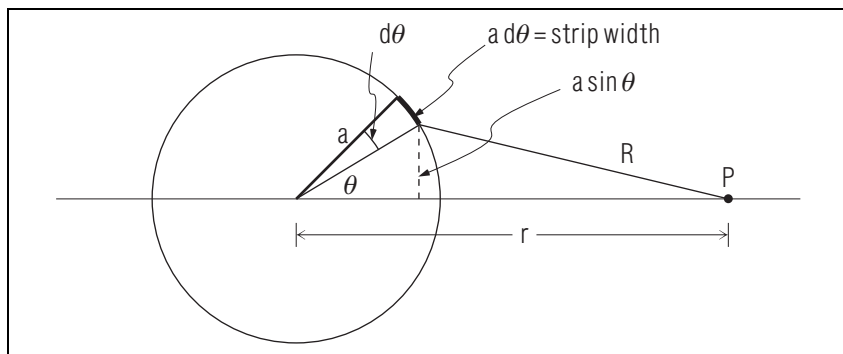
P. Signell and J. S. Kovacs

1. Introduction

One of the remarkable features of an inverse-square-law force field is that it is zero inside a spherical shell of sources, and that outside the shell it is exactly that which would be produced by a point source at the origin. The latter result is rigorously derived in this unit, using straightforward summation of the forces due to all sources. In turn, the field and potentials outside homogeneous spherical masses are derived and used to justify the method used elsewhere¹ for finding G from g .

2. Study Material

In AF² study Section 15.8 (p. 318-322), which contains the derivation. This sketch may be helpful:



Just as remarkable as the theorems proved above is the fact that the gravitational field is exactly zero inside a spherical mass shell. You may wish to show this also, while you are at it. However, a very simple, very elegant, but somewhat subtle proof of these theorems can be given in

¹In MISN-0-101.

²M. Alonso and E. J. Finn, *Physics*, Addison-Wesley, Reading (1970). For availability, see this module's *Local Guide*.

terms of Gauss's Law.³

Acknowledgments

Preparation of this module was supported in part by the National Science Foundation, Division of Science Education Development and Research, through Grant #SED 74-20088 to Michigan State University.

³See "Gauss's Law Applied to Spherically Symmetric Charge Distributions" (MISN-0-132).

LOCAL GUIDE

The readings for this unit are on reserve for you in the Physics-Astronomy Library, Room 230 in the Physics-Astronomy Building. Ask for them as “The readings for CBI Unit 109.” Do **not** ask for them by book title.